

Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$. If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x .

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \rightarrow x$, if it contains all open neighborhoods of x .

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

converges to x . This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.

- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \emptyset$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X .
- A function $f : X \rightarrow Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \rightarrow x \in X$ implies $f(F) \rightarrow f(x) \in Y$.

Definition 1.4. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\rightarrow \subset \mathcal{F}(X) \times X$. Then $(X, \rightarrow)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \rightarrow x$;
- If $F \rightarrow x$ and $G \rightarrow x$, then $F \cap G \rightarrow x$;
- If $F \rightarrow x$ and $F \subset G$, then $G \rightarrow x$.

Definition 1.5. Let X, Y be topological spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y .

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.